Project 1: Data Integrity and Cleaning Process Documentation

1. Data Cleaning & Preparation

- **Handling Missing Values:** Missing and null values were identified and processed using **Power Query**, ensuring data completeness and consistency across the dataset.
- **Removing Duplicates:** Duplicate records were detected and removed using Excel's **Remove Duplicates** feature to maintain data accuracy.
- **Data Formatting:** Date, numeric, and text fields were standardized to the appropriate formats, enabling reliable calculations and analysis.
- **Data Validation:** The dataset was reviewed for inconsistencies, and necessary corrections were applied to improve overall data quality.

2. Data Quality Verification

- **Record Validation:** The **COUNTA** function was used to verify the total number of records and ensure data integrity after cleaning.
- **Conditional Formatting:** Applied conditional formatting rules to identify potential anomalies, blank cells, duplicate values, and outliers for further review.
- **Format Consistency Check:** Data type mismatches (such as dates stored as text or numbers) were identified and corrected to ensure consistency throughout the dataset.

3. Outcome

- Successfully transformed raw data into a clean, structured, and analysis-ready dataset.
- Improved data accuracy, consistency, and reliability through systematic cleaning and validation procedures.
- The dataset is now fully prepared for exploratory analysis, reporting, and business insights generation.

4. Conclusion

Through the use of Power Query, data validation techniques, conditional formatting, and Excel-based quality checks, the dataset was successfully cleaned, standardized, and optimized for analysis. Missing values, duplicate records, formatting inconsistencies, and potential data quality issues were systematically identified and resolved to improve the overall reliability of the dataset.